

Solution STA510 Exam spring 2023

Problem 1

a)

Clearly $f(x) \geq 0$ for all $0 \leq x \leq 4$ as the square-root is positive for positive arguments. Then it remains to show that

$$\int_0^4 f(x)dx = \int_0^4 \frac{3\sqrt{x}}{16} dx = \frac{3}{16} \left[\frac{2}{3} x^{3/2} \right]_0^4 = \frac{3}{16} \frac{2}{3} 4^{3/2} = \frac{1}{8} 8 = 1$$

I.e. $f(x)$ is a proper probability density function.

b)

$$E(X) = \int_0^4 x \frac{3\sqrt{x}}{16} dx = \int_0^4 \frac{3x^{3/2}}{16} dx = \frac{3}{40} \left[x^{5/2} \right]_0^4 = \frac{12}{5} = 2.4.$$

c)

$$E(\sqrt{X}) = \int_0^4 \sqrt{x} \frac{3\sqrt{x}}{16} dx = \dots = \frac{3}{2}.$$

Further,

$$E((\sqrt{X})^2) = E(X) = \frac{12}{5}.$$

Hence

$$Var(\sqrt{X}) = \frac{12}{5} - \left(\frac{3}{2} \right)^2 = \frac{3}{20} = 0.15.$$

d)

The CDF is defined as

$$F(x) = \int_{-\infty}^x f(y)dy = \int_0^x \frac{3\sqrt{y}}{16} dy = \frac{3}{16} \left[\frac{2}{3} y^{3/2} \right]_0^x = \frac{x^{3/2}}{8}.$$

e)

First we need to find the inverse CDF:

$$F(x) = \frac{x^{3/2}}{8} = u \Leftrightarrow x^{3/2} = 8u \Leftrightarrow x = (8u)^{2/3} = 4u^{2/3} = F^{-1}(u).$$

In order to simulate n random numbers with density $f(x)$ one would

- simulate n uniform(0,1) random numbers, say U_i , $i = 1, \dots, n$.
- apply the inverse CDF to each of the uniform numbers, i.e. $X_i = 4U_i^{2/3}$, $i = 1, \dots, n$.

Then X_i , $i = 1, \dots, n$ will have density $f(x)$.

f)

The correct function is `p1.inv.3` which does exactly as indicated in point e).

- `p1.inv.1` simply plugs `uniform(0,1)` random numbers into the PDF, and will produce random numbers between 0 and $3/16$ (and not between 0 and 4 as it should).
- `p1.inv.2` plugs `uniform(0,4)` random numbers into the inverse CDF. This will lead to random numbers between 0 and $4 \times 4^{2/3} \approx 10.08$ (and not between 0 and 4 as it should).
- `p1.inv.4` plugs `uniform(0,1)` random numbers into the CDF, and will produce random numbers between 0 and $1/8$ (and not between 0 and 4 as it should).

g)

From a performance perspective, we wish to choose the proposal distribution that leads to the fewest rejected proposals. Hence we need to find the smallest $c_1 \geq f(x)/g_1(x)$ and $c_2 \geq f(x)/g_2(x)$.

Consider the uniform first:

$$f(x)/g_1(x) = 4 \frac{3\sqrt{x}}{16}$$

Since this function is monotonously increasing, we have that $f(x)/g_1(x) \leq f(4)/g_1(4) = 3/2 = c_1$.

Then the truncated Gamma proposal density:

$$f(x)/g_2(x) = \frac{1}{C\sqrt{x}\exp(-x/20)} \frac{3\sqrt{x}}{16} = \exp(x/20) \frac{3}{16C}$$

Also this function is monotonously increasing, hence $f(x)/g_2(x) \leq f(4)/g_2(4) = 3\exp(4/20)/(16C) = c_2 \approx 1.084782223$.

Since $c_2 < c_1$ we would prefer g_2 as the proposal distribution.

Problem 2

a)

Point estimate $\hat{A} = 4X$ where $X = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(u_i, v_i)$. We have a realization of X ($= 0.166846$), which gives the best estimate: $4 \times 0.166846 = 0.667384$.

CI: (using that X is an average) $SD(\hat{A}) = 4SD(X) = 4SD(\mathbf{1}(u_i, v_i))/\sqrt{n} \approx 4 \times 0.3728385/\sqrt{1000000} = 0.001491354$. Hence the (normal-based) CI becomes $[0.667384 - 1.96 * 0.001491354, 0.667384 + 1.96 * 0.001491354] = [0.6644609, 0.6703071]$.

b)

Since we have that $Y = \sum_{i=1}^n \mathbf{1}(u_i, v_i) \sim \text{Binomial}(\frac{1}{6}, n) \Rightarrow SD(Y) = \sqrt{n(1/6)(1-1/6)} = \sqrt{5n/36}$ and that $\hat{A} = \frac{4}{n}Y$ it follows that

$$SD(\hat{A}) = \frac{4}{n}SD(Y) = \frac{4\sqrt{5/36}}{\sqrt{n}}.$$

To find the number of simulations needed in order to make $SD(\hat{A}) < 0.01$ we solve the equation:

$$\frac{4\sqrt{5/36}}{\sqrt{n}} = \frac{1}{100} \Leftrightarrow \sqrt{n} = 400\sqrt{5/36} \Leftrightarrow n = (400\sqrt{5/36})^2 = 200000/9 \approx 22222.222222$$

I.e. we need at least 22223 simulations to make the simulation with $SD(\hat{A}) < 0.01$.

Sanity check, not required:

```
n <- 22223
nrep <- 1000
Ahats <- numeric(nrep)
for(i in 1:nrep){
  u <- runif(n,min=-1,max=1)
  v <- runif(n,min=-1,max=1)
  indicator <- (sqrt(abs(u))+sqrt(abs(v)))^2 <= 1.0
  Ahats[i] <- 4.0*mean(indicator)
}
print(sd(Ahats))

## [1] 0.009320456
```

c)

Since $\hat{A}$ involves a sum of random variables, we know from the CLT that $\hat{A}$ is approximately normally distributed, and the standard deviation of $\hat{A}$ was found above to be $\frac{4\sqrt{5/36}}{\sqrt{n}}$.

Then it only remains to find $E(\hat{A})$ to fully characterize the CLT:

$$E(\hat{A}) = \frac{4}{n}E(Y) = \frac{4}{n} \frac{n}{6} = \frac{2}{3}.$$

Hence $P(\hat{A} < 0.67) \approx P(Z < (0.67 - 2/3)/SD(\hat{A})) = P(Z < 0.07071068) \approx 0.5279$ where the latter number comes from the standard normal table.

Sanity check:

```
n <- 1000
nrep <- 1000
Ahats <- numeric(nrep)
for(i in 1:nrep){
  u <- runif(n,min=-1,max=1)
  v <- runif(n,min=-1,max=1)
  indicator <- (sqrt(abs(u))+sqrt(abs(v)))^2 <= 1.0
  Ahats[i] <- 4.0*mean(indicator)
}
print(mean(Ahats<0.67))

## [1] 0.548
```

Problem 3

a)

There are a multitude of different possibilities here

Alternative 1: Accept/reject based on $Uniform(-2, 2)$ proposal:

$$f(x)/g(x) = 4f(x) \leq 4f(0) = 4C \approx 1.671838 = c$$

The algorithm follows the steps:

- simulate $x \sim Uniform(-2, 2)$

- calculate $\alpha = f(x)/(c/4) = \exp(-x^2/2)$
- simulate $U \sim \text{Uniform}(0, 1)$.
- if $U < \alpha$ accept x as a random draw from $f(x)$, otherwise go back to the first step.

Alternative 2: Accept/reject with $N(0, 1)$ proposals (since the target density is a truncated standard normal), i.e. $g(x) = \exp(-x^2)/\sqrt{2\pi}$.

$$f(x)/g(x) = \begin{cases} \sqrt{2\pi}C & \text{if } -2 \leq x \leq 2, \\ 0 & \text{otherwise,} \end{cases} \leq \sqrt{2\pi}C \approx 1.047669 = c$$

Note now that the accept probability reduces to

$$f(x)/(cg(x)) = \begin{cases} \frac{C \exp(-x^2/2)}{\sqrt{2\pi}C \exp(-x^2/2)/\sqrt{2\pi}} & \text{if } -2 \leq x \leq 2, \\ 0 & \text{otherwise,} \end{cases} = \begin{cases} 1 & \text{if } -2 \leq x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

The AR-algorithm therefore reduces to

- simulate $x \sim N(0, 1)$
- if $-2 \leq x \leq 2$ accept x , otherwise go back to the previous step.

Notice It is also fine come directly to the above algorithm without involving the AR-step, but this requires knowledge of how to simulate from truncated distributions using random draws from the “un-truncated” distribution.

b)

This is a independent (proposal) Metropolis Hastings algorithm. The proposal distribution is $\text{Uniform}(-2, 2)$.

Problem 4

a)

$$\begin{aligned} A &= \int \exp\left(-\frac{x^2}{2}\right) \cos(x) dx = \int \frac{\exp\left(-\frac{x^2}{2}\right) \cos(x)}{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx \\ &= \sqrt{2\pi} \int \cos(x) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx. \end{aligned}$$

The latter integral may be approximated using the estimator

$$\hat{A} = \sqrt{2\pi} \frac{1}{n} \sum_{i=1}^n \cos(x_i), \quad x_i \sim \text{iid } N(0, 1).$$

Sanity check: true $A \approx 1.520346901$.

```
A.hat <- sqrt(2*pi)*mean(cos(rnorm(1000000)))
print(A.hat)
```

```
## [1] 1.520903
```

b)

Similar reasoning: $\mathbf{1}_{(a,b)}(x)$ is equal to 1 if $a < x < b$ and 0 otherwise:

$$B = \sqrt{2\pi} \int \mathbf{1}_{(-\pi/2, \pi/2)}(x) \cos(x) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

which suggest the MC estimator:

$$\hat{B} = \sqrt{2\pi} \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{(-\pi/2, \pi/2)}(x_i) \cos(x_i), \quad x_i \sim \text{iid } N(0, 1).$$

Sanity check: true $B \approx 1.630009116$.

```
x <- rnorm(1000000)
B.hat <- sqrt(2*pi)*mean(cos(x)*(x<pi/2)*(x>-pi/2))
print(B.hat)
```

```
## [1] 1.628813
```

c)

Crude MC estimator

$$\tilde{B} = \frac{\pi}{n} \sum_{i=1}^n \exp\left(-\frac{x_i^2}{2}\right) \cos(x_i), \quad x_i \sim \text{iid } \text{Uniform}(-\pi/2, \pi/2)$$

Sanity check: true $B \approx 1.630009116$.

```
x <- runif(1000000,min=-pi/2,max=pi/2)
B.hat <- pi*mean(cos(x)*exp(-0.5*x^2))
print(B.hat)
```

```
## [1] 1.629404
```

d)

Here

$$\begin{aligned} A &= \int \exp\left(-\frac{x^2}{2}\right) \cos(x) dx = \int \frac{\exp\left(-\frac{x^2}{2}\right) \cos(x)}{\exp(-|x|/2)/4} \exp(-|x|/2)/4 dx \\ &= 4 \int \exp(-x^2/2 + |x|/2) \cos(x) \exp(-|x|/2)/4 dx \end{aligned}$$

which may be estimated using

$$\tilde{A} = \frac{4}{n} \sum_{i=1}^n \exp(-x_i^2/2 + |x_i|/2) \cos(x_i), \quad x_i \sim \text{iid } \text{Laplace}(0, 2).$$

Sanity check: true $A \approx 1.520346901$.

```
n <- 1000000
x <- rexp(n,rate=0.5)*sample(c(-1,1),n,replace = TRUE)
print(4*mean(exp(-0.5*x^2+0.5*abs(x))*cos(x)))
```

```
## [1] 1.518786
```

e)

Similar to above

$$\tilde{B} = \frac{4}{n} \sum_{i=1}^n \mathbf{1}_{(-\pi/2, \pi/2)}(x_i) \exp(-x_i^2/2 + |x_i|/2) \cos(x_i), \quad x_i \sim \text{iid } \text{Laplace}(0, 2).$$

f)

Antithetic associated with $U_i \sim Uniform(-\pi/2, \pi/2)$ is simply $-U_i$. Hence antithetic estimator becomes

$$\bar{B} = \frac{\pi}{n} \sum_{i=1}^{n/2} \left[\exp\left(-\frac{x_i^2}{2}\right) \cos(x_i) + \exp\left(-\frac{(-x_i)^2}{2}\right) \cos(-x_i) \right], \quad x_i \sim \text{iid } Uniform(-\pi/2, \pi/2).$$

However, note that $\bar{B}$ may be rewritten (as $x^2 = (-x)^2$ and $\cos(x) = \cos(-x)$) as simply

$$\bar{B} = \frac{\pi}{n/2} \sum_{i=1}^{n/2} \exp\left(-\frac{x_i^2}{2}\right) \cos(x_i)$$

Hence we get the same estimator as in c), but with half the Monte Carlo sample size (but still at the cost of n function evaluations). I.e. In this case, with a symmetric integrand, Antithetic variables should not be used.

Problem 5

a)

This is Gibbs sampler, since the elements of the vector are updated according to the conditional distributions $p(x_1|x_2)$ and $p(x_2|x_1)$ (see lecture notes “chap11_part3”, page 6, for details).

b)

This is a Random Walk MH sampler with $N(x_t, I_2)$ proposal distribution for x_{t+1} . It is seen that the accept probability reduces to $\min(1, p(x_{proposed})/p(x_t))$, i.e. does not involve the proposal mechanism.

c)

This is iid simulation random vectors based on that $x_i = L\varepsilon, \varepsilon \sim N(0, I)$ and $LL^T = \Sigma$.

d)

iid simulation (i.e. the latter) is to be preferred in this case, as it produces un-correlated samples which are to be desired in most applications of stochastic simulation.